

Couch Potato Paradise: Smart Home System Using Sensor Data

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Introduction

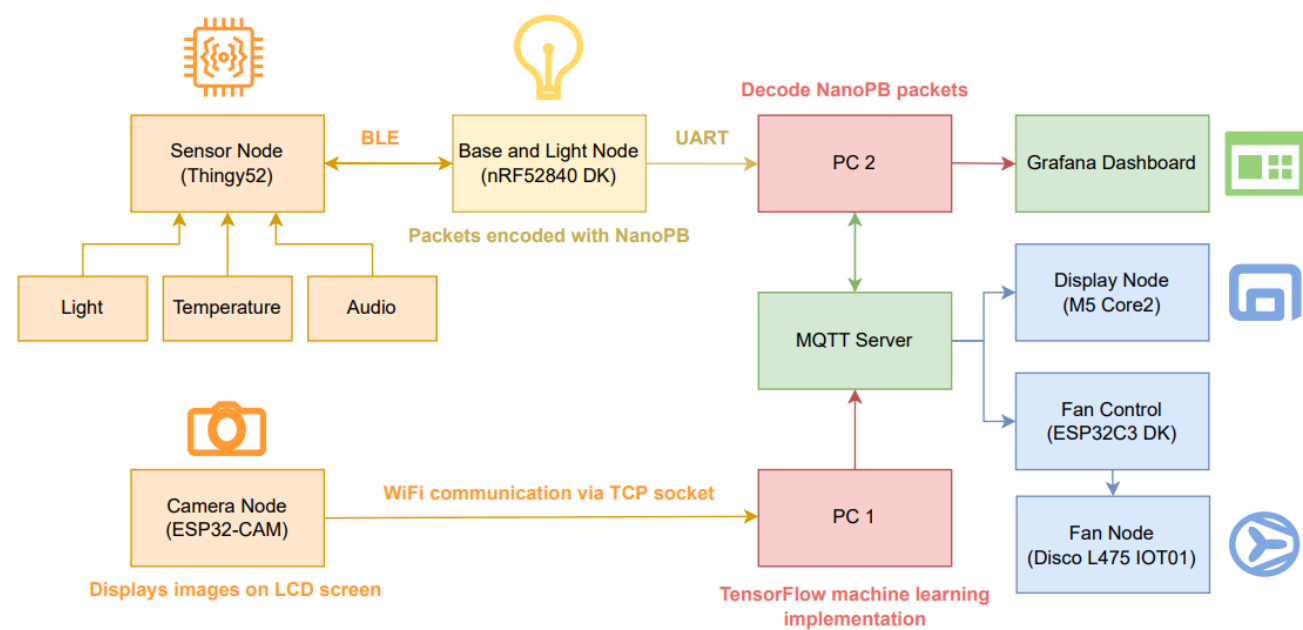
Developed using the Zephyr RTOS, Couch Potato Paradise is a multi-sensor embedded system that responds to user gestures, audio cues and environmental data in real time, to create a fully automated smart home system.

Couch Potato Paradise uses machine learning for image recognition, IoT techniques in the form of MQTT for PC-to-node communication and nanoPB for encoding sensor data.

Sensor nodes provide real-time environment data, displayed on a dashboard for users to see. Audio detection allows users to easily adjust the room's light level. Machine Learning removes the need for physical human interactions, classifying human hand images to change the output of fans (and air conditioners).

Combining BLE, UART, Wi-Fi and MQTT communication, a seamless flow of information is provided quickly and accurately between nine system nodes, three sensors and one actuator.

System Overview



- 1. Data Collection:** Temperature, light and clap status are transmitted via BLE from the Thingy to the nRF DK, where they are encoded using nanoPB. CLI commands can be used to send a “sensor on/off” message from the nRF DK to the Thingy via BLE. The encoded data is sent to PC 2. Image data is transmitted via WiFi from the Camera Node to PC 1.
- 2. PC:** PC 1 uses machine learning to classify the image and send the fan setting via MQTT. PC 2 decodes the nanoPB packet and receives the fan setting via MQTT from PC 1.
- 3. Viewers:** MQTT publishes the sensor values and fan speed to display on the M5 and dashboard.
- 4. Actuator:** MQTT publishes the fan speed to the ESP32C3, where GPIO output are provided to the Disco board which actuates the servo motor through PWM.

Aims (KPIs) → Findings

- 1. Machine Learning Accuracy**
Image Recognition algorithm must be able to correctly identify gestures with minimum 70% accuracy.
 - 2. Sensor Reading Accuracy**
Clap detection must achieve minimum 60% accuracy.
 - 3. Sensor Latency**
Temperature and light sensors able to take 2 measurements per second. PDM able to detect 2 distinct claps per second.
 - 4. Display Functionality**
M5 Core2 updated within 2 s of Image Classification. Grafana Dashboard: updated every 5 seconds
 - 5. Robustness**
Image and audio recognition remain reliable across different users and different environmental conditions (including windy and noisy conditions).
 - 6. Communication Latency**
ESP32 transmits image data within 5 s. PC reads image data, performs image recognition and communicates to M5 Core2 within 2 s. Thingy52 transmits sensors data and light setting via BLE within 0.5 s.
- 1. Machine Learning Accuracy**
The image recognition model achieved ~80% accuracy, in specific environments (limited background noise).
 - 2. Sensor Reading Accuracy**
The clap detection achieved ~80% accuracy.
 - 3. Sensor Latency**
All sensors met the required latency targets. For the final implementation, clap detection occurred constantly, and temperature and light measurements occurred every 2 s.
 - 4. Display Functionality**
The dashboard was successfully updated every 5 s. The M5 Core2 achieved ~3 s latency for all displays (temperature, light on/off and fan speed).
 - 5. Robustness**
Image recognition was highly reliable for different users. More data should be collected to improve reliability across different environments. Clap recognition worked in noisy conditions.
 - 6. Communication Latency**
Image data was transmitted in < 5 s. BLE communication occurred in < 0.5 s.

Limitations

The most significant limitation of the project was that the nRF boards (Thingy52 and DK) did not have WiFi modules. Allowing the Thingy52 to communicate directly with the MQTT server would have reduced the overall system complexity (number of nodes). Notably, the inability of the ESP32C3 to configure a PWM signal (tested and confirmed) meant that the disco board had to be employed as the Fan Node (servo driver).